

Machine Learning A (2026)

Home Assignment 2

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1 Preprocessing (35 points)

1.1 Importance of Preprocessing (8 points)

The nearest neighbor to C is B, so BoL should not give him credit according to the algorithm.

$$\begin{aligned} A : \sqrt{(21 - 47)^2 + (36 - 35)^2} &\approx 26.02 \\ B : \sqrt{(21 - 22)^2 + (36 - 40)^2} &\approx 4.12 \end{aligned}$$

If instead the income is measured in dollars, A would be the nearest neighbor, so BoL should give C credit.

$$\begin{aligned} A : \sqrt{(21 - 47)^2 + (36000 - 35000)^2} &\approx 1000.34 \\ B : \sqrt{(21 - 22)^2 + (36000 - 40000)^2} &\approx 4000.00 \end{aligned}$$

This is because Euclidean distance is not scale-invariant.

1.2 Input Centering (9 points)

1.2.1 (a)

$$\begin{aligned} z_n = x_n - \bar{x} &\implies Z = X - \mathbf{1}\bar{x}^\top \\ &= X - \mathbf{1} \left(\frac{1}{N} X^\top \mathbf{1} \right)^\top \\ &= X - \frac{1}{N} \mathbf{1}\mathbf{1}^\top X \\ &= \left(I - \frac{1}{N} \mathbf{1}\mathbf{1}^\top \right) X \\ &= \gamma X \end{aligned}$$

1.2.2 (b)

TODO

1.3 Input Whitening (18 points)

1.3.1 (a)

$$\begin{aligned} \text{Var}(x_1) &= \text{Var}(\hat{x}_1) = 1 \\ \text{Var}(x_2) &= \text{Var}(\sqrt{1 - \epsilon^2}\hat{x}_1 + \epsilon\hat{x}_2) \\ &= (1 - \epsilon^2)\text{Var}(\hat{x}_1) + \epsilon^2\text{Var}(\hat{x}_2) \\ &= (1 - \epsilon^2)(1) + \epsilon^2(1) \\ &= 1 \\ \text{Cov}(x_1, x_2) &= \text{Cov}(\hat{x}_1, \sqrt{1 - \epsilon^2}\hat{x}_1 + \epsilon\hat{x}_2) \\ &= \sqrt{1 - \epsilon^2}\text{Cov}(\hat{x}_1, \hat{x}_1) + \epsilon\text{Cov}(\hat{x}_1, \hat{x}_2) \\ &= \sqrt{1 - \epsilon^2}(1) + 0 \\ &= \sqrt{1 - \epsilon^2} \end{aligned}$$

1.3.2 (b)

$$\begin{aligned} \hat{x}_1 = x_1, \hat{x}_2 = \frac{x_2 - \sqrt{1 - \epsilon^2}x_1}{\epsilon} &\implies f(x) = \hat{w}_1(x_1) + \hat{w}_2\left(\frac{x_2 - \sqrt{1 - \epsilon^2}x_1}{\epsilon}\right) \\ &= \left(\hat{w}_1 - \hat{w}_2\frac{\sqrt{1 - \epsilon^2}}{\epsilon}\right)x_1 + \left(\frac{\hat{w}_2}{\epsilon}\right)x_2 \\ \implies w_1 &= \hat{w}_1 - \hat{w}_2\frac{\sqrt{1 - \epsilon^2}}{\epsilon}, \quad w_2 = \frac{\hat{w}_2}{\epsilon} \end{aligned}$$

1.3.3 (c)

$$f(\hat{x}) = \hat{x}_1 + \hat{x}_2 \implies \hat{w}_1 = 1, \hat{w}_2 = 1$$

Plugging these into our equations from section 1.3.2:

$$w_1 = 1 - \frac{\sqrt{1 - \epsilon^2}}{\epsilon}, \quad w_2 = \frac{1}{\epsilon}$$

The minimum C needed for the constraint $w_1^2 + w_2^2 \leq C$ is exactly $w_1^2 + w_2^2$:

$$\begin{aligned}
C_{min} &= \left(1 - \frac{\sqrt{1-\epsilon^2}}{\epsilon}\right)^2 + \left(\frac{1}{\epsilon}\right)^2 \\
&= 1 - 2\frac{\sqrt{1-\epsilon^2}}{\epsilon} + \frac{1-\epsilon^2}{\epsilon^2} + \frac{1}{\epsilon^2} \\
&= 1 - 2\frac{\sqrt{1-\epsilon^2}}{\epsilon} + \frac{2-\epsilon^2}{\epsilon^2} \\
&= 1 - 2\frac{\sqrt{1-\epsilon^2}}{\epsilon} + \frac{2}{\epsilon^2} - 1 \\
&= \frac{2}{\epsilon^2} - \frac{2\sqrt{1-\epsilon^2}}{\epsilon}
\end{aligned}$$

1.3.4 (d)

As $\epsilon \rightarrow 0$, the dominant term $\frac{2}{\epsilon^2}$ prevails and $C_{min} \rightarrow \infty$. This means that as inputs become highly correlated, the weights required to construct the target function become infinitely large.

2 Hoeffding's Bound (15 points)

$$\begin{aligned}
\mathbb{E}[X] &= \frac{-2 + 0.8 + 1}{3} = \frac{-0.2}{3} = -\frac{1}{15} \\
\mathbb{E}[S] &= N\mathbb{E}[X] = -\frac{100}{15} = -\frac{20}{3} \\
X &\in [a, b] \text{ with } a = -2, b = 1 \\
\mathbb{P}(S > 22) &\implies \mathbb{P}(S - \mathbb{E}[S] > 22 - \mathbb{E}[S]) \\
&= \mathbb{P}\left(S - \mathbb{E}[S] > \frac{86}{3}\right)
\end{aligned}$$

Using Hoeffding's Inequality for sums:

$$\mathbb{P}\left(S - \mathbb{E}[S] \geq \frac{86}{3}\right) \leq \exp\left(-\frac{2 \cdot \left(\frac{86}{3}\right)^2}{100 \cdot 3^2}\right) \approx 0.1610$$

3 Illustration of Markov's, Chebyshev's, and Hoeffding's Inequalities (25 points)

3.a

1.

2.

3.

4.

5.

6.

7.

3.b [optional, 0 points]